

EconECPE: A Span-Level Emotion-Cause Dataset for Financial Social Media

— Supplementary Material —

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S1. SAMPLING AND PROMPTS

Strata. The silver sample is drawn in one deterministic hash-ordered scan: 15% pure random, 42.5% topic-bucketed and 42.5% lexically emotive-scored, with per-year proportional quotas. The topic stratum uses 14 high-precision keyword buckets aligned one-to-one with the keyword-addressable cause categories (all 16 except INTERPERSONAL and OTHER; first-match order, lowercase substring match), with an equal per-bucket quota and a per-(bucket, year) cap so no bucket clumps into a single era. The emotive stratum scores each comment by a 44-term affect lexicon plus surface signals (exclamation marks, an affective-emoji set, and all-caps words of length > 3), and comments scoring ≥ 2 qualify. The bucket keyword lists and the lexicon are in the release.

Prompts. Every annotator receives the frozen specification as a system prompt, reproducing the emotion and cause taxonomies, an arousal-based tie-break rule, the verbatim-span requirement, and the tricky-case protocol (sarcasm, quoted or reported affect, rhetorical questions, and the interpersonal restriction), followed by five few-shot exemplars covering multi-pair extraction, a post-title cause, sarcasm, a neutral comment, and interpersonal affect. It returns the typed-pair JSON schema (an empty list for a neutral comment; at most eight pairs). A separate judge prompt, given only the disputed fields and the competing candidate values, resolves genuine ties, with an explicit irony re-read instruction for sarcasm disagreements. The prompts verbatim, the schema and all pipeline code are in the release.

S2. ANNOTATION QUALITY CONTROL

Pilot and harness. The 250-comment pilot comprised 40% top-level comments on link posts (stressing post-title causes), 40% replies from argumentative threads, and 20% random comments; 99.2% of its comments carried ≤ 2 pairs. First-pass JSON validity ranged 88.9-96.1% across the three annotators, lowest for MiniMax-M3, whose inputs are the harder comments.

Consensus. The targeted third vote was estimated at $\sim 44\%$ of comments on the pilot and measured at 57.2% at full scale; the judge adjudicated 25.8% of silver pairs.

Hand audits. Two audits back the dominance of *anger/disgust*: 126 pilot pairs plus 30 neutral calls (90-93% acceptable), then 40 gold comments over-sampling *anger/disgust*, sarcasm and neutral.

Null causes. UNCLEAR appears on 42 of the 30,977 proposed clusters and survives the support gate at 19% against 47.6% overall, leaving 8 corpus-wide and none on the random stratum: null-cause proposals are real but low-agreement.

Matching details. Normalization folds case, whitespace, markdown escaping and quote style and strips per-token edge punctuation; the validator’s separate verbatim check stays exact. IoU penalizes length (a short span nested in a longer one scores $|T_{\text{short}}|/|T_{\text{long}}|$), which is why the relaxed criterion uses containment $|T_a \cap T_b| / \min(|T_a|, |T_b|) \geq 0.8$. Cause-span accuracy is also reported *conditional* on a matched emotion span, separating cause localization from pair recall.

Gold adjudication funnel. Table S1 breaks down the 173 debated comments. The author review covered all 29 comments on which the debaters raised specification questions (46 questions): 23 outcomes were confirmed and 6 borderline cases decided by the authors; two gold records changed (one neutral comment re-labeled *pessimism/despair* at intensity 0.4, one cause re-categorized from GEOPOLITICS to OTHER). Recurring question themes were: whether a continuous rant is one pair or several; the affect threshold (whether calm condemnation, idioms or loaded vocabulary carry enough “heat” for *anger/disgust*); INTERPERSONAL vs. topic routing when anger targets another user’s economic claim; cause-category boundaries (e.g. HOUSING vs. PERSONAL FINANCE); and sincere vs. ironic readings.

TABLE S1
DEBATE ADJUDICATION OF THE 173 FLAGGED GOLD COMMENTS.

Converged at the first exchange (2 turns)	92
Converged in 3-6 turns	64
Converged after escalation to Grok-4.3	14
Split after re-adjudication (GLM-5.2), majority kept	3
Adopted: Claude Sonnet 5 / Gemini 3.5 Flash / joint list	66 / 36 / 68
Comments with spec questions (questions)	29 (46)
Author review: confirmed / decided / records changed	23 / 6 / 2
Total adjudication cost	\$12.38

Human benchmark procedure. Per comment the tool pooled every pair from the nine label sources, deduplicated on spans and typed fields, and ranked by backing count; the author kept correct candidates, discarded the rest, rewrote fields where a span was right but its labels were not, and marked the comment neutral where nothing held. Of 596 distinct candidates (median 6 per comment) it kept 109 and rejected 487 (82%), marked 12 comments neutral with candidates still

on the table, and edited 2 kept pairs, yielding 109 pairs at a 36% neutral rate against Claude Sonnet 5’s 110 and Gemini 3.5 Flash’s 85.

S3. MODEL TRAINING AND EXTRACTION DETAILS

Generative student. We train in bf16 rather than 4-bit QLoRA, an un-ablated precaution against quantization error in Qwen3.5’s recurrent linear-attention blocks. LoRA (rank 16, $\alpha = 16$, no dropout) is applied to all attention and MLP projections, with the loss masked to the assistant turn; AdamW-8bit, cosine schedule (peak 1×10^{-4}), effective batch 16, up to 3 epochs with dev-loss checkpoint selection. The 4,096-token window truncates nothing.

Encoder tagger. DeBERTaV3-large reads the concatenated context and comment (512 tokens) with three heads on the shared token representations: a typed emotion-span head (BIO, 15 classes), a typed cause-span head (BIO, 35 classes), and an intensity-regression head on emotion-span tokens. Silver character spans are aligned via the fast tokenizer’s offset mapping (92%/91% of emotion/cause spans recovered); it trains on the same split as the deployed student.

Further comparisons (bootstrap, 2,000 resamples). Consensus exceeds MiMo-V2.5, the strongest teacher on E-span, by $\Delta = +8.2/+7.9$ (E-span/+E) and $+7.9$ on +CS (all $p \leq 0.002$). The recall student is distinguishable from the support- ≥ 2 student ($\Delta = +6.8$, $p = 0.002$) but not from the consensus ($\Delta_{\text{E-span}} = +1.2$, $p = 0.67$; $\Delta_{\text{+CS}} = +2.1$, $p = 0.48$).

S4. MARKET STUDY: ROBUSTNESS DETAIL

Minimum- p Granger statistic. Taking the minimum p over lags 1-4 is an optimistic within-cell choice: under simulated independence it rejects at 11.3% for a nominal 5%. The bias makes significance easier to obtain, so it can only strengthen a negative finding; it is the positive findings that it inflates.

Further screening signals (raw). Euphoria reversal: Granger $p \approx 2$ to 4×10^{-4} . Fear share predicts an equity rebound ($r \approx +0.12$, Granger $p \approx 0.01$) in-sample only. *Anger* predicts next-week VIX changes (Granger $p \approx 10^{-4}$), consistent with a sentiment $\rightarrow$ volatility channel. The one cause-conditioned cell to reach the top of the ranking is fear $\times$ MONETARY POLICY $\rightarrow$ gold ($r = +0.13$), not Granger-significant ($p = 0.12$). Fear share co-moves with same-week VIX at $r = +0.10$ ($p = 0.04$).

Within-regime construct validity. Demeaning within each of the five macro regimes (boundaries 2020-01, 2021-07, 2023-01, 2024-01) leaves both VIX couplings essentially unchanged, with the expected sign in four of five, though the individual windows (52-131 weeks) are underpowered; prediction *within* regimes is left to future work for the same reason.

Out-of-sample selection. Selection uses in-sample data only; across the 15 emotion features (14 plus net valence) the euphoria correlation is not family-wise significant (block-bootstrap max-statistic $p = 0.40$), but it holds on the disjoint window. The fear rebound does not ($r = +0.01$, $p = 0.47$): it is carried by the 2020 crash-recovery, and a block bootstrap

concurrs that euphoria is autocorrelation-robust where the fear effect is fragile.

Sarcasm. Valence is aggregated on the intended emotion, so we re-run the screen under *drop* (sarcastic pairs excluded, $-343,455$) and *flip* (emotion kept, valence sign inverted). Exposure is structurally small: 21.9% of *anger/disgust* pairs carry the flag against 2.1% of *excitement/euphoria* and 0.7% of *fear* (14.9% overall). The euphoria reversal moves from $r = -0.144$ to -0.147 under drop and is exactly unchanged under flip, per-emotion features being invariant to valence sign by construction. The VIX coupling, the one claim resting on signed valence, survives its worst case ($r = -0.157 \rightarrow -0.153, -0.145$).

External baselines. Raw correlations with our net valence are large (FinBERT $r = 0.95$, LM $r = 0.89$ in differences) but mostly shared posting volume; partialling volume out leaves $r = 0.21$ (LM), 0.31 (VADER), 0.12 (FinBERT) and -0.21 for LM %negative, all $p < 0.02$. The strongest external screening result is VADER tone $\rightarrow$ 1-week equity ($r = -0.115$, raw $p = 0.015$), the same reversal sign as euphoria.

Time-varying censoring. The weekly removed-or-deleted rate rises with market fear (Spearman $\rho = 0.59$ with the VIX, $n = 444$) and is dominated by topic/quality moderation rather than user deletion ($\sim 6.5:1$). Reweighting each week by its surviving-text share and dropping the heavily moderated 2022 peak, separately and together, leave the euphoria reversal intact (out-of-sample r between -0.17 and -0.18 , one-sided $p \leq 0.012$) and the fear $\times$ INFLATION arm null. Detection depends on when the archive last re-scraped a comment, so the rate is itself a lower bound.

Affect-enriched training distribution. The student trains on a sample only 15% of which is uniform, so it over-predicts pairs on ordinary comments, inflating the *level* of every weekly series. This is benign for the market features, which are shares or first differences, but the corpus-wide pair rate is not a prevalence estimate.

Label noise and the bound. Split-half reliability of the weekly series (Spearman-Brown corrected) is 0.99 for aggregate net valence and 0.89 and 0.53 for the fear $\times$ GEOPOLITICS and fear $\times$ INFLATION cells, the latter sparse. Disattenuation rescales the null distribution by the same $\sqrt{\rho_{xx}}$ as the estimate, so p -values are unchanged and the flagship stays null (min $q = 0.31$); correcting the upper end of the 95% interval for the largest flagship correlation gives $|r| \approx 0.13$ as the largest true cause-conditioned correlation consistent with the data.

Second community. Pooling *r/economics* and *r/investing* (4,392,861 pairs) roughly doubles the pairs per week, not the ~ 446 independent weeks. On the pooled corpus fear share co-moves with the VIX at $r = +0.19$ and net valence at $r = -0.30$; the pooled flagship is null (fear $\times$ INFLATION $\rightarrow$ gold $r = -0.00$, $p = 0.93$; family min $q = 0.12$); and *r/investing*’s weak momentum sign cancels the euphoria reversal out of sample.